DERIVABLE JUDGEMENT

A Statistical Decision-Making Model
Part A – Theoretical Foundation

1. What is inferential statistics?

Inferential statistics is used to study a sample and make conclusions about a larger population. It helps us estimate values and make decisions using sample data.

2. What is hypothesis testing and its components?

Hypothesis testing is a method used to check a claim with the help of sample data. Its main components are the null hypothesis (H■), alternative hypothesis (H■), significance level, test statistic, p-value, and final decision.

3. Explain confidence interval and critical value.

A confidence interval is a range of values used to estimate the actual value of a population parameter. A critical value is a cutoff value used to decide whether the null hypothesis should be rejected.

4. Define p-value.

A p-value tells us how likely the observed result, or a more extreme result, would be if the null hypothesis were true. If the p-value is less than 0.05, we usually reject the null hypothesis.

5. Differentiate Type I and Type II errors.

A Type I error happens when we reject a null hypothesis that is actually true. A Type II error happens when we do not reject a null hypothesis that is actually false.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

Z-test: It is used to test means or proportions when the required conditions for a z-test are satisfied.
T-test: It is used to compare the means of two groups, especially when the population standard deviation is unknown.
Chi-square test: It is used to check whether two categorical variables are related.
ANOVA test: It is used to compare the means of three or more groups.

7. What is Covariance?

Covariance shows how two numerical variables change together. Positive covariance means they tend to move in the same direction, while negative covariance means they tend to move in opposite directions.

8. What is Correlation?

Correlation shows the strength and direction of the relationship between two variables. Its value ranges from -1 to +1. A value near +1 shows a positive relationship, a value near -1 shows a negative relationship, and a value near 0 shows little or no linear relationship.

Part A – Completed